

Chapter 4: Multimodal Retrieval Guide

Visual-BGE & Milvus Cross-Modal Vector Search Architecture

This subfolder provides the complete implementation for Multimodal RAG & Visual Retrieval. It leverages BAAI's Visualized_BGE (a joint text-image embedding model) paired with Milvus vector database to enable seamless cross-modal search—allowing users to search image databases using text queries (Chinese & English) or image-to-image similarity search.

1. Subfolder File Inventory & Purpose

File Name	Type / Format	Core Purpose & Functionality
.env.example (Top-level & Nested)	Config Template	Defines environment variables including Milvus host/port, Visualized_BGE model weights path, and HuggingFace cache directories.
requirements.txt	Dependencies	Lists PyPI libraries required for multimodal encoding and vector database storage: pymilvus, torch, transformers, FlagEmbedding, pillow, and matplotlib.
Milvus+Visual-BGE-MultimodalRetrieval-Chinese.py	Python Script	Cross-modal retrieval pipeline for Chinese text queries & image targets. Maps Chinese text and game screenshot images into a unified 768d/1024d embedding space using Visualized_BGE, populates Milvus, and executes cross-modal similarity search.
Milvus+Visual-BGE-MultimodalRetrieval-English.py	Python Script	Cross-modal retrieval pipeline tailored for English text queries & image datasets using Visualized_base_en_v1.5.pth. Demonstrates text-to-image and image-to-image vector matching in Milvus.
Milvus+Visual-BGE-PureRetrievalProgram.py	Python Script	Standalone pure retrieval engine. Connects to an existing populated Milvus collection to execute instant text-to-image or image-to-image queries without rebuilding the index.
search_results.jpg	Image Artifact	Rendered visual search output generated by matplotlib showing the input query on the left and top-K visually matching image results on the right with cosine similarity scores.
search_with_filter.jpg	Image Artifact	Visualization of multimodal vector search combined with scalar metadata filtering (e.g. category == 'weapon'), ensuring retrieved images satisfy exact business rules.
search_without_filter.jpg	Image Artifact	Visualization of pure unconstrained vector similarity search results, serving as a baseline comparison against scalar-filtered search.

2. Technical Mechanics & Code Architecture

A. Visualized_BGE Joint Embedding Space

Standard embedding models only accept text strings. Visualized_BGE extends BGE models with a visual encoder (e.g., EVA-CLIP / ViT) that projects both image inputs (`PIL.Image.open()`) and text queries into the exact same vector dimension. Because text and images share a coordinate space, the dot product between a text vector and an image vector directly measures visual-semantic relevance.

Snippet 1: Dual-Modal Vector Generation

```
# Joint Encoding with Visualized_BGE
from FlagEmbedding import Visualized_BGE

model = Visualized_BGE(model_name_bge="BAAI/bge-base-zh-v1.5",
                        model_weight="./Visualized_base_en_v1.5.pth")

# Encode Image to Vector
image_vector = model.encode(image_path="scene_01.jpg")

# Encode Text Query to Shared Vector Space
text_vector = model.encode(text="A warrior standing in a snowy mountain forest")
```

B. Milvus Schema for Multimodal Records

The Milvus collection schema defines fields for storing image embeddings alongside scalar metadata for filtering:

Snippet 2: Milvus Multimodal Schema

```
from pymilvus import FieldSchema, CollectionSchema, DataType, Collection

fields = [
    FieldSchema(name="id", dtype=DataType.INT64, is_primary=True, auto_id=True),
    FieldSchema(name="image_path", dtype=DataType.VARCHAR, max_length=512),
    FieldSchema(name="category", dtype=DataType.VARCHAR, max_length=64),
    FieldSchema(name="image_vector", dtype=DataType.FLOAT_VECTOR, dim=768)
]
schema = CollectionSchema(fields, description="Multimodal Image Search Collection")
collection = Collection("multimodal_demo", schema)
```

C. Cross-Modal Querying with Metadata Filtering

When executing a search, the text embedding is passed to `collection.search()`. Adding an `expr` parameter (e.g. `expr='category == "boss_battle"'`) restricts similarity search to specific image categories, preventing irrelevant matches even if visual features are similar.

WHY MULTIMODAL RETRIEVAL MATTERS

Multimodal vector search enables searching visual assets (product images, PDF diagrams, video frames) using natural language queries without relying on manual text tags or alt-text descriptions.

3. Comparison of Chapter 4 Subfolders

Subfolder Name	Primary Retrieval Modality	Key Technologies Used	Primary Use Case
HybridRetrieval	Text (Dense + Sparse)	Milvus, BGE-M3 (1024d Dense + Lexical Sparse), Cross-Encoder Reranker	Enterprise document search requiring high keyword recall and semantic precision.
LlamaIndex	Text / Persistence	LlamaIndex, StorageContext, Local File JSON Serialization	Lightweight local RAG without running external database server instances.
MultimodalRetrieval	Cross-Modal (Image + Text)	Milvus, Visualized_BGE (Joint Image-Text Embedding), PyTorch, Matplotlib	Visual asset management, e-commerce image search, diagrammatic PDF page retrieval.